MMA301_SU26_PE_20260701
Subject: MMA301
Duration: 85 minutes

Thông tin chung

INSTRUCTIONS

Read the instructions carefully before you begin the questions.

- You may not use any device to share data with others.
- You must use an IDE such as Visual Studio Code as your development tool.
- You must use NPM and Expo for building your application.
- All fruit data must come from the fruits.json file and be stored in MockAPI.io.
- Screen switching must be responsive (no noticeable lag).
- Your StudentCode is your roll number (e.g., se123456).

IMPORTANT – Before you start building your application, you MUST complete the following steps:

1. Review your work before submitting it to make sure there are no syntax or compilation errors in the submitted PE.
2. The data for this application must be retrieved from the fruits.json file and stored on MockAPI.io. The endpoint must be added to your environment variables in the .env file. If you fail to do so, your score will be 0.
3. If your PE program contains any syntax or compilation errors, it will not meet the requirements, and your mark will be 0.
4. Create a new folder named MMA301PE_SU26_StudentCode.
5. If your code includes anything unrelated to the test, your work will be considered invalid (0 points).

Task 1. Build a new React Native application.

1. (0.5 marks) Create a new React Native application named after your **StudentCode** (e.g., se123456). The app must include **three** bottom tabs, and all screens must be switchable with a single touch.
2. (0.25 marks) Create a resource on MockAPI.io to store **fruit** data. **The resource name must be your StudentCode.**

Task 2. Implement the actions on the Home screen.

1. (1.75 marks) Build the Home screen in your application, and allow filtering fruits by selecting an **origin** on this screen. The data should include:
   - **fruitName**
   - **imageUrl**
   - **ripeness** (displayed as a percentage, not a decimal value)
   - **isSeedless must be visually highlighted**, e.g., shown with one icon for true (seedless) and another icon for false (seeded).

   The list must be retrieved from MockAPI.io, filtered by **origin**, and displayed in descending order of **storageTime**.

   Use Bottom Tabs to switch between the three screens: **Home**, **Favorite**, **Explore**.

   Design and lay out this screen as neatly and efficiently as possible.
2. (0.25 marks) On the **Home** screen, tapping the fruit image or the fruitName title should navigate to the **Detail** screen.
3. (1.0 marks) On the **Home** screen, implement the favorite list with AsyncStorage. When the favorite icon on a fruit is pressed, the fruit should be added to the favorite list.

Task 3. Implement the actions on the Detail screen.

1. (1.75 marks) On the Detail screen, show all fruit information:
   - **fruitName**
   - **ripeness** (displayed as a percentage)
   - **imageUrl** (rendered as image)
   - **sweetness** (Low, Medium, High)
   - **storageTime**: convert from days into weeks and days using these rules:

*— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —*

```
- If both weeks and days exist, show as "X week(s) Y day(s)" (e.g., 10 → "1 week 3 days").
- If only weeks exist, show as "X week(s)" (e.g., 14 → "2 weeks").
- If only days exist, show as "X day(s)" (e.g., 5 → "5 days").
```

- **colors** (shown as badges or tags for multiple colors)
- **origin**

- **isSeedless** (styled same as on **Home** screen)

Additionally:

- Display the favorite state
- Allow toggling the favorite state (add/remove from Favorite list)

Design and lay out this screen as neatly and efficiently as possible.

Task 4. Implement some actions at the Favorite screen.

1. (0.5 marks) On the **Favorite** screen, show the list of **fruits** stored in **AsyncStorage**.
2. (0.75 marks) Allow changing the favorite state of individual fruits or of all fruits (e.g., remove a single fruit or clear the entire list) by confirming a notification. The favorite list on the screen should update after the action is completed successfully.
3. (0.25 marks) Tapping a model's image should navigate to the **Detail** screen.

Task 5. Implement some actions at the Explore screen.

1. (2.5 marks) On the **Explore** screen, show a list of **fruits** that meet the following conditions:
    - Exclude all fruits with origin "**Vietnam**".
    - Storage time must be **between 5 days and 6 weeks**.
    - Exclude all fruits that have **Green** in their colors list.
    - After filtering, sort the results:
        - First, by **ripeness ascending** (low to high).
        - If two fruits have the same ripeness, sort by **fruitName descending** (Z to A).
2. (0.25 marks) Allow changing the favorite state of individual fruits with a confirmation dialog. The favorite list on the screen should update after the action is completed successfully.
3. (0.25 marks) Tapping a fruit's image or the fruitName title should navigate to the **Detail** screen.

-The end of test-